

BAYANKHONGOR, Mongolia

WMO: 442870

Lat: **46.1739N**Lon: **100.7103E**Elev: **6099**StdP: **11.73**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-16.5	-30.2	1.3	-14.3	-27.0	1.5	-11.9	22.8	-1.7	20.5	5.3	7.4	0	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	82.8	57.5	79.7	56.3	76.8	55.2	61.9	75.9	59.7	73.5	57.7	70.9	9.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.1	87.6	67.4	54.6	79.6	64.9	52.4	73.6	63.8	30.9	75.9	29.2	73.6	27.7	71.0	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	25.1	21.2	19.0	DB	-24.2	87.5	4.0	2.3	-27.1	89.2	-29.4	90.5	-31.7	91.8	-34.6	93.4
(5)				WB	-24.7	63.1	3.9	4.1	-27.6	66.1	-29.9	68.5	-32.1	70.8	-34.9	73.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	0.2	7.3	21.7	38.6	49.6	60.5	65.1	61.7	50.3	34.4	17.1	5.1	(6)
(7)		DBStd	24.00	8.52	9.65	10.71	8.71	8.09	6.21	5.58	5.40	7.44	8.73	9.98	8.20	(7)
(8)		HDD50	7026	1544	1195	877	348	106	5	0	2	85	483	988	1393	(8)
(9)		HDD65	11279	2009	1615	1342	791	478	159	70	129	443	947	1438	1858	(9)
(10)		CDD50	1347	0	0	0	7	95	320	467	365	93	0	0	0	(10)
(11)		CDD65	124	0	0	0	0	1	23	72	27	1	0	0	0	(11)
(12)		CDH74	1291	0	0	0	0	37	303	662	276	13	0	0	0	(12)
(13)		CDH80	241	0	0	0	0	3	52	142	44	0	0	0	0	(13)
(14)	Wind	WSAvg	8.0	7.2	7.5	8.3	9.2	9.5	8.6	7.5	7.4	7.5	7.3	7.5	7.7	(14)
(15)	Precipitation	PrecAvg	8.2	0.1	0.1	0.2	0.4	0.6	1.5	2.3	1.9	0.7	0.3	0.1	0.1	(15)
(16)		PrecMax	14.1	0.4	0.4	1.4	1.5	2.8	4.1	6.0	4.8	2.1	2.1	0.8	0.4	(16)
(17)		PrecMin	4.3	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.9	0.1	0.1	0.3	0.4	0.7	1.1	1.2	1.3	0.5	0.4	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.5	35.3	53.1	67.1	78.4	85.4	87.6	84.5	75.8	61.5	46.3	29.0	(19)
(20)			MCWB	20.6	26.3	37.7	46.5	52.0	57.7	61.9	57.1	52.5	43.4	34.4	22.2	(20)
(21)		2%	DB	21.0	30.1	47.6	62.4	73.3	80.8	83.9	80.4	71.1	56.9	40.7	24.4	(21)
(22)			MCWB	16.4	22.9	33.4	42.9	50.0	56.3	58.7	56.3	50.2	41.2	30.8	18.8	(22)
(23)		5%	DB	16.5	26.0	42.8	58.6	69.3	77.5	80.8	76.9	67.4	52.8	35.8	20.4	(23)
(24)			MCWB	12.8	20.0	31.0	41.1	48.1	54.8	57.4	55.5	48.7	38.7	27.6	15.8	(24)
(25)		10%	DB	12.3	21.7	38.1	54.6	65.8	74.1	77.7	73.7	63.7	48.6	31.3	16.8	(25)
(26)			MCWB	9.3	17.0	28.4	38.9	46.4	53.3	56.3	54.5	47.1	36.3	24.8	13.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	26.4	38.5	49.7	56.1	63.0	66.6	63.5	56.2	46.2	34.8	22.5	(27)
(28)			MCDWB	26.3	33.8	51.0	64.9	74.8	79.0	82.7	76.1	71.0	59.1	45.6	27.9	(28)
(29)		2%	WB	16.5	23.2	34.2	44.7	52.4	59.2	62.8	60.6	52.8	42.2	30.9	19.0	(29)
(30)			MCDWB	20.8	29.7	46.0	59.0	69.3	74.6	77.4	73.3	64.9	55.0	39.9	24.1	(30)
(31)		5%	WB	12.8	20.2	31.4	41.9	49.9	57.1	60.6	58.5	50.5	39.5	27.8	16.0	(31)
(32)			MCDWB	16.4	25.9	42.0	56.4	65.8	72.0	75.2	70.0	63.8	51.4	35.3	20.2	(32)
(33)		10%	WB	9.4	17.0	28.7	39.4	47.5	55.3	58.5	56.5	48.4	36.8	24.9	13.2	(33)
(34)			MCDWB	12.3	21.7	38.1	53.6	63.5	69.0	72.3	68.2	61.4	47.6	31.1	16.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.7	21.7	22.5	22.0	22.2	21.2	19.5	20.0	21.2	21.2	19.6	17.6	(35)
(36)			MCDBR	21.3	24.6	25.7	27.1	26.9	25.6	23.9	24.2	24.6	25.1	22.6	20.4	(36)
(37)			MCWBR	17.9	19.2	16.2	14.8	13.3	11.0	8.8	9.5	11.6	14.7	15.6	16.5	(37)
(38)		5% WB	MCDWB	21.0	24.3	24.6	25.5	24.9	22.1	20.8	19.0	22.2	24.0	22.2	20.2	(38)
(39)			MCWBR	17.8	19.1	16.0	15.5	14.3	11.6	10.3	10.2	12.0	15.1	15.8	16.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.224	0.251	0.293	0.333	0.348	0.359	0.357	0.333	0.301	0.273	0.242	0.225	(40)
(41)		taud		2.402	2.338	2.267	2.231	2.248	2.308	2.361	2.403	2.444	2.448	2.430	2.396	(41)
(42)		Ebn at Noon		289	298	297	292	291	287	286	290	292	286	278	276	(42)
(43)		Edh at Noon		22	29	36	41	42	40	37	34	30	25	21	19	(43)
(44)	All-Sky Solar Radiation	RadAvg		633	977	1443	1890	2166	2137	2028	1840	1590	1107	698	525	(44)
(45)		RadStd		51	55	58	82	135	133	104	103	71	52	41	35	(45)